

Big Data Analytics : Project Assignment

Objective

The aim of this project is to allow students to apply the skills learned in the course to a real-world dataset. Students will collect, clean, analyze, and present insights based on the data, using appropriate analytical and visualization techniques.

Project Requirements

1. Choose a Topic

Students may select a dataset and topic of their choice, but it must be approved by the instructor. Example domains include:

- Sports performance analytics
- Education analytics
- Environmental data analysis
- Social media sentiment analysis
- Financial market trends
- Healthcare and wellness insights

2. Dataset

- Minimum 500 rows of data.
- Data may be collected from public datasets, APIs, web scraping, or personal collection.
- Clearly cite the source of the data.

3. Tasks to Complete

- *Data Collection*: Find and document the dataset source. Explain the data structure (columns, data types, meaning).
- *Data Cleaning & Preprocessing*: Handle missing values, duplicates, and incorrect formats. Apply data transformations if needed (e.g., normalization, feature engineering).
- *Exploratory Data Analysis (EDA)*: Summarize data using descriptive statistics. Visualize patterns using at least 5

different chart types (bar chart, line chart, scatter plot, heatmap, etc.).

- *Analysis / Modeling*: Use at least one statistical or machine learning technique (e.g., correlation analysis, regression, clustering, classification).
- *Insights & Conclusion*: Provide clear answers to at least 3 research questions. Summarize key findings and possible recommendations.
- *Presentation*: Submit both a written report (PDF or DOCX) and a presentation file (PowerPoint, Google Slides). Present your findings in a 10–15 minute session.

Deliverables

Report (5–10 pages) including:

1. Title & group members
2. Introduction & objectives
3. Data description & source
4. Data cleaning process
5. Exploratory Data Analysis (with charts)
6. Analytical methods used
7. Results & discussion
8. Conclusion & recommendations
9. References

Presentation summarizing key insights.

Code notebook (.ipynb, .py, or R script) with comments.

Evaluation Criteria (Total 100 points)

Criteria	Points
Dataset quality & relevance	10
Data cleaning & preprocessing	15
EDA & visualization	20
Analytical/modeling methods	20
Insights & conclusions	15
Report clarity & structure	10
Presentation quality	10

Important Dates

Final submission & presentation: Week 12